

Chapter 85 — Benign Disorders of the Penis and Urethra

Campbell-Walsh-Wein Urology · Chapter Notes

OVERVIEW OF RECONSTRUCTIVE SURGICAL TECHNIQUES

- **Goal of genitourinary reconstruction:** a safe, functional and aesthetically acceptable urinary tract, which requires attention to **vascular supply, minimal tissue trauma and the dynamics of urinary flow**.
- **Reconstructive versus extirpative surgery:** reconstructive results depend on methods that **minimise tissue damage and maximise wound healing** — adequate visualisation, appropriate suture, delicate handling, correct positioning and adequate retraction.

Visualisation and cautery.

- **Loupes and headlamps** are widely used; **avoid non-neutral neck positions**, as long-term use is associated with work-related musculoskeletal injury.
- **Bipolar cautery on the urethra** — active and return electrodes are both in the handpiece, so damage to mucosa and surrounding spongiosum is minimised. **Monopolar on subcutaneous tissue and muscle.**

Instruments.

- A **plastic surgery tray** usually suffices: fine tenotomy and Metzenbaum scissors (kept sharp to limit collateral trauma), fine forceps, skin hooks, and needle holders delicate enough for fine tissue yet firm enough to prevent the needle moving.
- **Toothed forceps** on muscle, fascia and subcutaneous fat; **non-toothed (DeBakey) forceps** on the urethra.
- **Bougie à boule sizers** — **calibration** of the urethral lumen (essential). **Goodwin sounds** — **dilation, not calibration**; safer than van Buren sounds because they carry a lumen for a guidewire. Either may be passed through a suprapubic tract into the bladder neck during posterior urethroplasty.

Sutures.

- **Always absorbable** in urethral reconstruction — permanent suture is a **nidus for stone**. Preferred: **monofilament polydioxanone**; braided absorbable is acceptable.
- Wound support is needed for **about 30–60 days**; **chromic gives only 14 days → not recommended** for urethral work.
- Calibre 4-0 to 6-0: **5-0 for most urethral work**; **6-0 for single-layer mucosal anastomosis** — non-transecting excision and primary anastomosis (EPA) with mucosectomy, or the inner ventral layer of a transecting EPA. Needle: **small tapered (RB-1)**.

Position and retraction.

- Lithotomy or high lithotomy when exposure demands it, but **limit time → neuropathy and compartment syndrome**; move to **low lithotomy whenever possible** (for example during graft harvest). **Supine for penile surgery** when perineal access is not required.
- **Self-retaining perineal Bookwalter retractor** (authors prefer a segmented oblong Magrina ring with suture slots); alternatives are a Lone Star retractor with stay hooks or a modified Denis-Browne retractor slit for stay hooks.

KEY POINTS: RECONSTRUCTIVE SURGICAL TECHNIQUES

- **Planning:** imaging adequate to understand the problem, plus **several backup plans** if the primary approach is not feasible.

- Excellent visualisation (positioning and retraction), delicate handling and **fine absorbable suture**.

Concept — *Reconstruction succeeds or fails on the wound bed rather than on removal of disease, so every choice of instrument, suture, cautery and position is made to preserve blood supply and limit trauma.*

PRINCIPLES OF RECONSTRUCTIVE SURGERY

Tissue Transfer

- Tissues used: **skin, mucosa, muscle, peritoneum and omentum**; chosen by colour, texture, thickness, extensibility, innate tension and blood supply, which vary between individuals.
- **Graft versus flap**: a **graft** is separated from its blood supply; a **flap** is transferred with it. Both require knowledge of donor and recipient anatomy.
- All tissue has **extensibility, inherent tension and the viscoelastic properties of stress relaxation and creep**. These derive mainly from the **helical arrangement of collagen with elastin cross-linkages**, suspended in a mucopolysaccharide matrix that governs viscoelasticity.
- **Buccal mucosa suits the urethra** because its **thick lamina propria** makes it easier to handle, holds sutures better and is less prone to sacculation than skin.

Grafts

- A graft acquires a new blood supply from the host bed by **take** (Fig. 85.1), which requires **about 96 hours** in two phases, and is **impaired by anything that impairs host-bed vascularity**.

Phase of take	Duration	Mechanism	Graft temperature
Imbibition	~48 hours	Graft "drinks" nutrients from the adjacent host bed	Below core
Inosculation	~48 hours	True microcirculation re-established	Rises to core

Skin Grafts

- **Epidermis**: thin, water-impermeable, multilayered keratinised epithelium. **Dermis**: tough fibrous connective tissue carrying follicles, vessels, glands, lymphatics and nerves — **papillary dermis** (vessels, nerves, some stability) over **reticular dermis** (additional strength).

SPLIT-THICKNESS VERSUS FULL-THICKNESS SKIN GRAFT

Feature	Split-thickness skin graft (STSG)	Full-thickness skin graft (FTSG)
Layers carried	Epidermis + variable dermis; reticular dermis not carried	Epithelium + superficial and deep dermis; subdermal plexus exposed
Harvest	Dermatome, 0.015–0.05 cm (pneumatic dermatome 0.012–0.020 inch in buried penis repair)	Blade; ellipse from loose skin for primary closure; defat after harvest
Vascular plexus	Superficial plexus of small, numerous vessels → favourable take	Larger, sparse vessels → complex nutritional needs (fastidious)
Contraction, durability	Contracts; brittle, less durable	Contracts less; more durable
Lymphatics, physical characteristics	Few lymphatics; physical characteristics not carried	Most lymphatics and physical characteristics carried
Adnexa	Generally no hair follicles or sweat glands	Hair transferred — must be hairless where it meets urine
Size, donor site	Can be larger; donor re-epithelialises quickly, needs no closure	Limited by primary closure

Feature	Split-thickness skin graft (STSG)	Full-thickness skin graft (FTSG)
Cosmesis	Inferior	Superior

- **Meshing** an STSG expands it **up to sixfold**, covers irregular surfaces smoothly and lets fluid escape; an **unmeshed** graft can be "**pie-crust**" against haematoma and seroma, and gives **better long-term cosmesis**.
- **Hair**: skin that will contact urine or become inaccessible (urethra, apex of a neovaginal canal) **must be hairless**; pre-harvest laser or electrolysis may be used. Otherwise-discarded skin (scrotum at vaginoplasty; abdominal skin at panniculectomy and buried penis repair) can supply an FTSG.
- **Genital versus extragenital FTSG**: extragenital skin has **greater mass** → **more fastidious**, and the poor urethral results with extragenital FTSGs are attributed to **poor or ischaemic take**.
- **Exception — posterior auricular (Wolfe) graft**: thin skin over temporalis fascia on numerous perforators; its subdermal plexus mimics an intradermal plexus and its **mass resembles an STSG**.
- **Immobilise skin grafts 5–10 days** during revascularisation — bolster, or vacuum device over petroleum gauze.

Dermal Graft, Tunica Vaginalis Graft, Vein Grafts and Other Grafts

Graft	Properties	Use and limitations
Dermal	Exposes intradermal and deep dermal plexus; takes readily (not fastidious) ; physical characteristics of normal skin	Long used to augment tunica albuginea in Peyronie disease (PD)
Tunica vaginalis	Properly prepared, essentially peritoneum ; pliable, uniform thickness	Easy harvest, minimal donor morbidity
Vein	Not a true graft in this terminology: survives by direct endothelial perfusion and re-established flow through the vasa vasorum	Tunical defects in PD (long-term behaviour inferred from vascular literature). Saphenous vein for long urethral strictures: tubular shape helps, but valves and perfusion prevent wide acceptance
Bladder epithelial	Plentiful, distensible, highly vascular, accustomed to urine ; less easily harvested	Mini-laparotomy (with open suprapubic tube) or endoscopic laser harvest
Rectal mucosa	Long segments, minimal morbidity	Panurethral disease or when oral mucosa unavailable; small favourable series, more data needed; transanal endoscopic microsurgical or single-port robotic harvest

Oral Mucosa Graft

- Oral mucosa grafts (OMGs): **nonkeratinised** epithelium, **rich in elastin** (pliable, structurally sound), over a **highly vascular lamina propria** that promotes take; **panlamina plexus**; accustomed to a **wet environment** and resist infection (Fig. 85.1B).
- **Can be thinned**, provided enough **deep lamina** is kept to preserve physical and vascular characteristics.

Oral mucosa site	Location	Donor-site profile
Buccal mucosa graft (BMG)	Over buccinator, cheek	More swelling (first 30 days) and numbness (up to 6 months) than lingual
Lingual	Undersurface of tongue	More difficulty speaking and with tongue protrusion

Oral mucosa site	Location	Donor-site profile
Labial	Inner lower lip	Numbness, salivary change and pain worse than inner cheek ; little urethral data

- **Buccal and lingual give equal urethroplasty success** (meta-analysis of comparative studies); the sites differ only in donor morbidity. Grafts vary in thickness and substance.
- **Buccal onlay grafts are durable** into the long term, with **failures occurring within the first 12 months**.

Flaps

- A **flap** is excised and transferred with its blood supply **preserved or surgically re-established**; classified by **composition, blood supply, movement and location** (Figs. 85.2–85.4).
- **Axial flap**: a specific named vessel (or several named pedicles, as in gracilis) running in the flap's long axis; **indirect** if perforators pass through deeper tissue before the fascia, **direct** if they pass through fascia only. **Random flap**: no defined cuticular vascular territory.
- Location: **local** (defect adjacent to harvest site), **regional** (same region, not contiguous), **distant — pedicled** (still on native supply) or **free** (detached and anastomosed to recipient vessels, usually microvascular).

TABLE 85.1 FLAP CATEGORIZATION AND CHARACTERIZATION

Category	Subtype	Example
Composition	Fasciocutaneous	Distal penile circular fasciocutaneous flap
	Superficial fascia	Dartos flap
	Musculocutaneous (skin, fat and muscle)	Vertical rectus abdominis myocutaneous (VRAM) flap
	Muscle only	Gracilis flap
	Visceral	Omental flap
Blood supply	Axial — indirect (perforators through deeper tissue before fascia)	Distal penile circular fasciocutaneous flap; gracilis (axial, multiple pedicles)
	Axial — direct (perforators through fascia only)	—
	Random (no defined cuticular territory)	U-flap in perineal urethrostomy; Z-plasty
Movement	Advancement	Y-V advancement flap; U-flap in perineal urethrostomy
	Rotational	Martius flap
	Transposition	Z-plasty
Location	Local	Scrotal flap for a Fournier defect
	Distant — pedicled	Anterolateral thigh flap
	Distant — free	Radial forearm free flap phalloplasty

Modified from Buckley J, McAninch J, BJU Int 2007.

- **The categories are independent axes**: one flap sits in several rows — the U-flap is random *and* advancement; the distal penile circular fasciocutaneous flap is fasciocutaneous *and* axial-indirect; Z-plasty is random *and* transposition.
- **Trap**: Z-plasty is a **random** flap; gracilis is **axial**.

FLAP VASCULAR PATTERNS (FIGS. 85.2 AND 85.3)

Flap	Vascular continuity	Cutaneous continuity at base
Random	Arterial perforators interrupted; survives on intradermal and subdermal plexuses	Intact
Axial peninsula	Intact	Intact
Axial island	Intact (unsupported, "dangling" pedicle)	Divided
Microvascular free	Divided; reconstituted by microsurgical anastomosis	Divided

- Axial flaps survive on the large vessels entering the base **plus random distal vascularity**.
- **Musculocutaneous flap** (Fig. 85.4A): perforators from the artery to a muscle vascularise overlying fat and skin; can be free but usually transferred locally on its pedicle. **Fasciocutaneous flap** (Fig. 85.4B): perforators from plexuses on both surfaces of the fascia feed the paddle; in genital reconstruction these are **dartos-based penile flaps or forearm free flaps**.

KEY POINTS: PRINCIPLES OF RECONSTRUCTIVE SURGERY

- Grafts used successfully for **primary urethral reconstruction**: FTSG, OMG, bladder epithelial graft and rectal mucosal graft; the OMG has the most favourable vascular properties.
- FTSGs and STSGs for **penile reconstruction**; **microvascular free transfer** is a mainstay in complex cases; **dartos-based skin islands** for urethral reconstruction; dermal graft for tunical augmentation.

Concept — A graft must win a new blood supply and a flap brings its own. Every graft property follows from one trade-off: **dense fine plexus and low mass favour take**, while **carried dermis or lamina confers durability and resists contraction**. This explains STSG against FTSG, the failure of extragenital FTSG, the Wolfe exception, and why the OMG, with a panlamellar plexus, can be thinned.

SURGICAL ANATOMY OF THE PENIS AND MALE PERINEUM**Penile Anatomy**

- The shaft comprises **three erectile bodies** — **two corpora cavernosa and the corpus spongiosum** — with their fascia, nerves and vessels, covered by skin; all continue into the perineum.
- **Corpora cavernosa**: erectile tissue within a dense elastic sheath, the **tunica albuginea** (outer longitudinal and inner circular layers). They form a **single space** communicating through an **incompetent midline septum** that becomes more complete towards the base; the septum is strands interweaving with the tunica dorsally and ventrally (Fig. 85.5). Erectile tissue — arteries, nerves, muscle fibres and endothelium-lined venous sinuses (a sponge-like cut surface) — is separated from the tunica by thin areolar tissue.
- **Corpus spongiosum**: lies in the **ventral groove**; its tunica is **thinner** and it holds **less erectile tissue**; carries the urethra, expands distally as the **glans** (vertical slit meatus on the ventral tip) and broadens proximally between the diverging crura as the **bulb**.
- The base is supported by **two ligaments of mainly elastic fibres** continuous with the penile fascia (see Colles Fascia).
- **Layers, deep to superficial**: tunica albuginea → **Buck's fascia** → **dartos fascia** → skin.
- **Buck's fascia** — tough, elastic, applied to the tunica. Dorsally it encloses the **deep dorsal vein, paired dorsal arteries and dorsal nerve branches**; ventrally it **splits around the spongiosum**, and

consolidation lateral to it fixes the spongiosum to the corporal tunica. Distally it attaches to the **undersurface of the glans at the corona**; proximally it encloses each crus and the bulb, fixing them to pubis, ischium and inferior fascia of the perineal membrane.

- **Skin and dartos** — shaft skin is confluent with the glabrous glans skin and folds at the corona to form the **prepuce**. Dartos is areolar and **notably fat-free**, loosely attached to skin and Buck's fascia, carries the **superficial arteries, veins and nerves**, and fuses with Colles fascia in the perineum.

POSITION OF THE URETHRA IN THE ANTERIOR URETHRAL SEGMENTS (FIG. 85.6)

Level	Urethra within spongiosum	Corpora cavernosa
Bulbar	Eccentric (dorsal) ; lies against the triangular ligament	Split into separate crura
Penile shaft	Central	Fused, separated only by septal fibres
Coronal margin	Central; spongiosum now forms deep glans	Fused, septal fibres
Fossa navicularis	Widens ; relatively ventral ; wholly surrounded by glans tissue	—

Urethral Anatomy

- Divided into **anterior and posterior** by the **urogenital diaphragm (perineal membrane)**, and into **seven segments** (Fig. 85.8 numbers the six below the meatus).

Segment	Relations	Epithelium
Meatus	Vertical slit, slightly ventral of the tip	—
Fossa navicularis	Within the spongy tissue of the glans	Stratified squamous
Penile (pendulous)	Centre of spongiosum, distal to the ischiocavernosus investment, proximal to fossa	Pseudostratified columnar
Bulbar	Covered by bulbospongiosus (described by the authors as midline fusion of the ischiocavernosus); between membranous and penile; wider lumen , nearer the dorsal surface of spongiosum, exiting dorsally before the posterior attachment of bulbospongiosus to the perineal body	Squamous distally → transitional as it swings upward
Membranous	Traverses perineal membrane; surrounded by external sphincter; not attached to fixed structures; the only segment not invested by another structure	Delicate transitional
Prostatic	Surrounded by prostatic stroma and glands	Continuous with trigone and bladder
Bladder neck	Bladder neck musculature, variably surrounded by intravesical prostate	Continuous with trigone and bladder

- **Anterior urethra** = fossa, pendulous, bulbar (invested by spongiosum); **posterior urethra** = membranous, prostatic, bladder neck.

Sphincteric Anatomy

Sphincter	Site	Muscle	Control
Internal urinary sphincter	Bladder neck	Smooth (continuation of detrusor)	Autonomic, involuntary

Sphincter	Site	Muscle	Control
External sphincter (rhabdosphincter)	Membranous urethra	Striated	Voluntary

- Fig. 85.13 depicts the posterior urethral mechanism as internal sphincter, smooth muscle, intrinsic sphincter, external sphincter and an outer **volitional muscle of recruitment**.

Blood Supply

- **Penile skin:** left and right **superficial external pudendal vessels**, from the first part of the **femoral artery**, cross the upper medial femoral triangle and divide into **dorsolateral and ventrolateral branches** with collateralisation across the midline. Fine branches form a **rich subdermal plexus that sustains the skin after its dartos has been mobilised** — the basis of the **distal penile circular fasciocutaneous (McAninch) flap**. Accompanying veins are more prominent than the arteries.
- **Scrotal wall and ventral penile skin:** the **posterior scrotal artery**, a superficial branch of the deep internal pudendal artery, arborising in the tunica dartos and Colles fascia (Fig. 85.9B); supports the **longitudinal ventral penile skin (Turner-Warwick) flap**. The perineal artery continues lateral to the groin crease onto the thigh.

Arterial System

- **Deep structures:** internal iliac → internal pudendal → after the perineal branch it becomes the **common penile artery** along the medial inferior pubic ramus → near the bulb divides into **three terminal branches** (Fig. 85.10).

Branch	Course	Supplies and notes
Bulbourethral	Short, relatively large calibre; pierces Buck's fascia into bulbospongiosus; almost parallel to the membranous urethra	Bulb and spongiosum
Dorsal	Between deep dorsal vein (medial) and dorsal nerves (lateral) ; coiled, uncoils with erection to maintain flow	3–10 circumflex (circumflex cavernosal) branches with the circumflex veins supply the spongiosum; terminal arborisation in glans ; variable perforators to cavernosal arteries
Cavernosal	Enters corpus at the hilum , runs the shaft length	Helicine arteries (arterial erectile apparatus); often branches before entry; may cross to the opposite corpus; occasionally one artery supplies both sides

Venous Drainage

System	Drains	Route
Superficial	Penile skin and prepuce	Superficial dorsal vein in dartos (superficial to Buck's, parallel to deep dorsal vein) → saphenous vein, usually left
Intermediate	Spongiosum and distal two-thirds of corpora cavernosa	Circumflex veins (from spongiosum, around shaft, entering perpendicularly) and emissary veins (from sinuses through tunica, sometimes via circumflex) → deep dorsal vein (deep to Buck's, midline groove) → under pubis → prostatic (Santorini) plexus
Deep	Proximal third	Cavernous and crural veins → internal pudendal vein through Alcock canal → internal iliac vein

Lymphatics

Source	Nodes
Penile skin	Superficial inguinal
Glans	Collects at frenulum, circles dorsum of corona beneath Buck's fascia → bilateral deep inguinal and external iliac
Erectile tissue	Internal iliac

Nerve Supply

- **Pudendal nerve (S2–4)** → **dorsal nerves** branch in **Alcock canal**, pass through the perineal muscles and run **lateral to the dorsal arteries; sensory to penile skin and glans**. The pudendal nerve gives somatic motor and sensory innervation.
- **Cavernous nerves: sympathetic (T11–L2) and parasympathetic (S2–4)** fibres, with visceral afferents, via the **inferior hypogastric (pelvic) plexus** → along the **posterolateral border of the prostate** → branches to the sphincter → corpora. **Parasympathetic is proerectile; sympathetic is inhibitory**.

Perineum

- **Diamond-shaped outlet**: anteriorly the pubic arch and arcuate ligament; posteriorly the tip of the coccyx; laterally the inferior pubic and ischial rami. A line between the **ischial tuberosities** divides it into an **anterior urogenital triangle** and a **posterior anal triangle** (Fig. 85.12).

Colles Fascia

- Attached posteriorly to the **perineal body**, curves below the superficial transverse perineal muscles and projects forward as **two layers** attached laterally to the ischium and inferior pubic ramus.
- **Superficial fatty layer** → continuous with scrotal dartos, which contains **muscle fibres (scrotal rugae)** and projects, without muscle, as the scrotal septum. The **median raphe** runs from scrotum along the ventral penis and back along the perineum to the anus.
- **Deep membranous layer** → roofs the scrotal cavity, separating it from the superficial perineal pouch; joins scrotal dartos anteriorly, folds back beneath bulbospongiosus, and is continuous with penile dartos at the base.
- **Suspensory ligaments** (thickenings at the base): **outer fundiform ligament** — from the lower linea alba, splits around the penis and unites beneath; **inner triangular suspensory ligament** — from the anterior symphysis, blending with penile dartos.
- **Attachments**: anteriorly → **Scarpa fascia**; laterally → pubic arch and **fascia lata**; posteriorly → beneath the transverse perineal muscles to the **posterior perineal membrane**. The space beneath is the **superficial perineal pouch**, which confines infection, urinary extravasation and blood after urethral trauma.

SPREAD OF BLOOD, URINE AND INFECTION (FIG. 85.7)

Situation	Spread
Buck's fascia intact (penile fracture)	Confined to the shaft → eggplant deformity
Buck's fascia ruptured (straddle injury)	Through the dartos–Colles–Scarpa–tunica dartos system: scrotum, anterior abdominal wall, back to perineal body ; limited on the thigh by the fascia lata insertion
Fournier gangrene	Perineum or scrotum → anterior abdominal wall via Scarpa fascia

Superficial Perineal Space

- Lies **superficial to the perineal membrane and deep to Colles fascia**; contains the crura, proximal spongiosum and urethra, their muscles, and branches of the internal pudendal vessels and pudendal nerves.
- **Ischiocavernosus** covers the crura: from the inner ischium and ischial tuberosities, inserting into Buck's fascia around the crura below the arcuate ligament.
- **Bulbospongiosus** (midline fusion): attached to the perineal body posteriorly and to its fellow in the midline, encompassing bulb and crura; confluent laterally with ischiocavernosus and inserting into Buck's fascia over the dorsal vessels and nerves.
- **Deep perineal (Gallaudet) fascia** covers these muscles and is **continuous with Buck's fascia**.

Central Perineal Tendon (Perineal Body)

- Just **anterior to the anus** in the plane between the triangles; the interconnection of **eight perineal muscles**: anterior external anal sphincter fibres, **superficial transverse perineals** (from the ischial tuberosities, central insertion), posterior bulbospongiosus fibres, and, on its deep aspect, the deep transverse perineals and anterior levator ani.

Deep Perineal Space

- **Superior to the perineal membrane**; contains the urogenital diaphragm and deep transverse perineal muscles, **external urethral sphincter**, **bulbourethral (Cowper) glands**, vessels and nerves (Fig. 85.12C, D).
- **Sphincter fibres** arise from the medial inferior pubic rami and meet their fellows; in males they **encircle the membranous urethra** as its somatic sphincter.

KEY POINTS: ANATOMY OF THE PENIS AND MALE PERINEUM

- **Buck's fascia belongs to the deep structures**; the more areolar dartos fascia belongs to the skin and its vessels.
- **Genital skin** is supplied by the perineal–scrotal branch of the deep internal pudendal artery and by the **superficial external pudendal vessels** from the femoral arteries; **deep structures** by the common penile artery.

Concept — Two fascial envelopes define two vascular worlds: **dartos carries the skin's supply** (external pudendal and posterior scrotal), so skin flaps are raised on dartos, whereas **Buck's fascia encloses the erectile bodies** fed by the common penile artery. Which envelope is breached determines where blood, urine or infection travels.

SELECTED PROCESSES

Urethral Hemangioma

- **Rare but usually persistent**; presents with **haematuria or bloody urethral discharge**, occasionally obstruction; single or multiple; **the meatus is a common site**.
- **Cystoscopy** diagnoses it (dilated vessels), but the lesion **often extends beyond what is seen** → **preoperative magnetic resonance imaging (MRI)**.
- **All reported cases benign** → management by size and location: **asymptomatic** → **observe** (may regress spontaneously); **symptomatic** → **complete excision** to prevent recurrence.

Option	Role
Electrofulguration	Control of an acute episode only
Laser — argon, potassium titanyl phosphate (KTP, 532 nm), neodymium-doped yttrium aluminium garnet (Nd:YAG)	Smaller lesions ; successful with less scarring
Open excision and urethral reconstruction	Preferred for larger lesions , sometimes circumferential. Avoid tubed grafts ; tubed flap or mixed tissue transfer possible, staged reconstruction probably preferable
Sclerosing agents (e.g. triamcinolone)	Good initial success reported

Reactive Arthritis

- **Sterile synovitis after acute gastrointestinal or genitourinary infection**; the classic triad — **arthritis, conjunctivitis, urethritis** — occurs in **a minority**. Formerly **Reiter syndrome**, renamed because of Hans Reiter's association with Nazism.
- Onset **2–3 weeks** after *Chlamydia trachomatis*, human immunodeficiency virus (HIV), *Shigella* or *Salmonella* infection; **HLA-B27 in about two-thirds**.
- **Urethral involvement is usually mild and self-limited**.
- **Circinate balanitis** in up to half: small vesicles or **painless hyperkeratotic plaques with serpiginous borders** on the glans.

Lichen Sclerosus

- Lichen sclerosus (LS), formerly **balanitis xerotica obliterans (BXO)**: chronic inflammatory skin disorder, mainly genital; in males the **prepuce, glans and/or urethra**.
- **Epidemiology**: 1 in 300 to 1 in 1000 in Western populations; **more common in females**; male peak **30–50 years** but seen from infancy to old age; commonly found at **circumcision for phimosis beyond the neonatal period**.
- **Lesions**: erythematous → **pale, hypopigmented, sclerotic plaques**; ulceration, peeling, thickening. **The meatus is most commonly affected** and may be the early stage of later, more severe stricture from chronic obstruction. **Phimosis** in the uncircumcised (oedematous, thickened prepuce, possibly adherent to glans). Pain, pruritus, burning, obstructive voiding. **Chronic and typically progressive**.
- **Diagnosis is clinical; biopsy if inconclusive: band-like lymphocytic infiltrate, hyperkeratosis, epidermal atrophy, dermal oedema**. LS is regarded as **possibly premalignant**.

Proposed cause	Evidence
Autoimmunity	Autoantibodies to extracellular matrix; associations with thyroid disease, alopecia, vitiligo
Micro-RNA	LS strictures express different inflammatory and immune micro-RNAs from non-LS strictures
Genetics	Familial clustering; familial and autoimmune links stronger in females
Infection	Bacterial and viral triggers proposed; no definitive link
Trauma	Proposed
Chronic urine irritation	Post-void dribbling into foreskin or underwear, excess adiposity — certainly increases risk

Medical management.

- **Topical steroids** benefit many, though efficacy data are largely from females. Common regimen: **clobetasol propionate or betamethasone 0.05%, once or twice daily for 1–2 months, then alternate days for 3–4 months.**
- Urethral involvement: **intermittent catheterisation or meatal dilator lubricated with clobetasol** — for poor surgical candidates, the elderly, and younger patients with stable disease.
- **Phimosis only → circumcision may be curative.**

Meatal and urethral disease.

- Meatal stenosis: chronic dilation or surgery. Because a **proximal stricture often coexists, perform retrograde urethrography (RUG) first.**
- Surgery: meatotomy or **BMG onlay**. Restenosis is commoner than in non-LS stenosis → **extended meatotomy**; ventral or dorsal meatotomy with an **inverted-V relaxing incision** prevents puckering.
- Stricture mechanism unclear: repeated instrumentation, or **pressure voiding behind meatal stenosis → intravasation of urine into the glands of Littre** (Fig. 85.14). **Urethral LS is limited to the anterior urethra.**
- **Early meatal and fossa disease → prompt reconstruction** may avert panurethral stricture. Longer segments are reconstructable but challenging; counsel that **recurrence after urethroplasty is higher than in idiopathic or traumatic stricture.**
- **Staged versus single stage:** many advise staged BMG for all LS strictures, but a meta-analysis supports **single-stage repair for many. Use oral mucosa, never genital skin** — LS is a disease of genital skin.

Buried penis in LS.

- Shaft skin lost to inflammation traps the penis in the penopubic and scrotal area; patients are often **overweight** with prior circumcision or recircumcision. Management follows the **patient's wishes and functional need:**
- **Perineal urethrostomy** — technically straightforward because **LS spares the proximal anterior urethra**; or complete urethral reconstruction; or, for standing voiding and sexual function, **formal buried penis repair with skin graft.**
- Stricture plus buried penis → **staged:** limited release and urethroplasty first, formal repair with graft later. **Urethroplasty via penile invagination** avoids complications of penile skin incision. Counsel on **reburying and persistent glans LS.**
- **Malignant transformation in up to 8% of males: squamous cell carcinoma commonest;** verrucous carcinoma, basal cell carcinoma and melanoma also reported.

Concept — *LS is a progressive disease of genital skin that spares the proximal urethra. That single fact explains why the meatus is the sentinel site, why a RUG must precede meatal surgery, why oral mucosa rather than skin is used, why recurrence is high and why perineal urethrostomy is easy.*

Amyloidosis

- **Exceedingly rare** cause of obstructive voiding; also haematuria, dysuria, discharge or palpable mass — **consider in any urethral mass.** Workup: **cystoscopy and biopsy.** Management is symptom-directed: minimal symptoms → observe; bothersome obstruction → dilation or urethroplasty. **Progression and recurrence are rare.**

Urethrocutaneous Fistula

- **Epithelium-lined tract from urethra to skin; most commonly iatrogenic** — hypospadias repair, excision of urethral lesions, urethroplasty in gender-affirming phalloplasty — also piercings and periurethral infection with inflammatory stricture. **Treat the underlying process as well as the defect.**

Urethral Diverticulum: Male

- **Urothelium-lined outpouching** of the urethral lumen. **Congenital: all layers of the urethral wall; acquired: lacks smooth muscle.**
- **Acquired:** upstream of a stricture, after urethral surgery or chronic catheterisation, infection or trauma. A "congenital" **anterior** diverticulum may reflect incomplete urethral development or straddle trauma with intraspongiosal haematoma.
- Presentation: asymptomatic, **post-void dribbling**, recurrent urinary tract infection (UTI), mass, pain, **stone in the pouch**. Workup: **cystourethroscopy and RUG**.
- **Prostatic urethral diverticulum:** a large **müllerian remnant** with diminished virilisation, or often an **enlarged utricle in proximal hypospadias**. **Missed on voiding urethrography; shown by cystoscopy or RUG.**
- Consequences: the **catheter tip can enter the utricle** and drain a little urine, **falsely suggesting bladder placement**; large utricles fill on voiding and empty afterwards → **recurrent UTI and "incontinence"**.

Paraphimosis, Balanitis, and Phimosis

- **Paraphimosis:** painful swelling of foreskin **trapped proximal to the corona** → venous congestion → progressive oedema → **glans necrosis — a urological emergency**. Typical in **hospitalised patients after Foley catheterisation** when the foreskin is not replaced; also children and adolescents.
- **Reduction:** **steady pressure for several minutes** to express oedema → both thumbs push the glans while the fingers pull the foreskin forward. **Never a sudden traumatic reduction**. Granulated sugar (osmotic) has been suggested.
- Recurrence is common → **elective dorsal slit at minimum, or circumcision**. Irreducible after hours to days → **emergency dorsal slit or circumcision**; marked postoperative oedema is the rule.
- **Phimosis** (Greek "muzzling"): inability to retract the foreskin; **physiological in infants, resolving between 1 and 8 years**. In adults it prevents hygiene → smegma, balanitis, **penile cancer**. Treatment: topical steroid, dorsal slit, circumcision.
- **Balanitis** (Greek "acorn"): nonspecific glans inflammation, **far commoner in the uncircumcised**; from inadequate or, rarely, excessive hygiene. Smegma alone inflames or harbours organisms — **Candida commonest**; also *Streptococcus*, sexually transmitted infection (STI), chemical dermatitis and **plasma cell (Zoon) balanitis**. **Balanoposthitis** = glans plus foreskin.
- Treatment: keep clean and dry, exclude or treat STI, topical or oral antifungal for *Candida*, and **check blood glucose for undiagnosed or poorly controlled diabetes**. Balanitis **raises penile cancer risk**, so prevent recurrence.

Urethral Meatal Stenosis

- **Children:** complication of **newborn circumcision** (incidence quoted from 0.01% to 20%) — see Chapter 55.
- **Adults:** after **instrumentation or catheterisation**, or with **LS or hypospadias**.
- **Simple stenosis** → **ventral meatotomy** with interrupted sutures at the apex and on each side to appose tissue and control bleeding. Alternatively **chronic self-dilation with topical steroid as lubricant**, especially in LS. Extension into the fossa → urethroplasty (Chapter 86).

Circumcision

- Adult circumcision under local and/or general anaesthesia; **penile block along the dorsal nerves at the base**. For men and older boys the authors favour a **sleeve circumcision**:
1. Foreskin retracted → mark an incision leaving a **small preputial cuff**, straight across the frenular base → incise through dartos to the superficial lamina of Buck's fascia.
 2. Foreskin advanced → mark a second incision along the **coronal margin and the V of the frenulum**. Optional **frenuloplasty** (close the V longitudinally) makes the ventral skin straight, so the incision need not follow the V.
 3. Incise, **fulgurating vessels with bipolar cautery** as the edge is mobilised (adult vessels do not seal with compression) → remove the preputial sleeve → haemostasis.
 4. **Dartos with interrupted polyglactin (Vicryl) in each quadrant** → skin closure.
- **Complications uncommon**: temporary **glans hyperaesthesia** (resolves as glans skin keratinises); haematoma or minor dehiscence; minor cosmetic imperfections.
 - **Too much skin removed** → **discomfort with erection and buried penis**; **too little** → **adhesions and skin bridges**. Prevent by marking on **undistorted shaft skin**, involving the patient in marking, and psychosexual counselling.
 - **HIV prevention**: the landmark sub-Saharan African randomised controlled trials (RCTs) (Auvert 2005, Bailey 2007, Gray 2007) showed **about 60% reduction in female-to-male HIV acquisition**; United States retrospective data show a similar reduction among exposed heterosexual men.
 - Also reduces **herpes simplex virus type 2, human papillomavirus, genital ulcer disease and some bacterial STIs**.
 - **Men who have sex with men (MSM)**: meta-analysis shows reduced odds of HIV; protection is **greatest for predominantly insertive anal intercourse**, where an RCT found no seroconversions after circumcision against seroconversions in controls.
 - **Biological basis**: foreskin rich in **HIV-susceptible CD4+ T cells**, an anaerobic microbiome and inflammatory cytokines; removal lessens targets and inflammation. Uncircumcised men have more **genital ulcers and coital microtears**.

KEY POINTS: SELECTED PROCESSES

- Urethral haemangioma is benign and persistent; management depends on size and location.
- LS may be premalignant and causes phimosis, meatal stenosis and anterior stricture; results of LS stricture repair are often suboptimal and treatment follows the patient's goals.
- Urethrocutaneous fistula: treat the underlying process, not only the defect.

CURVATURES OF THE PENIS

- **Rigidity requires symmetric compliance**: as the penis engorges, the tunica albuginea and septal fibres stretch to their compliance limit and tumescence becomes rigidity. **Curvature = asymmetry** — diminished compliance or foreshortening of one aspect. Congenital and acquired curvature may look alike but differ in mechanism.

Feature	Congenital	Acquired
Age	Teens and young adults (more erections, sexual activity)	Largely older men
Direction	Most often ventral ; also dorsal, lateral	Dorsal, lateral, ventral or complex

Feature	Congenital	Acquired
Pain	Painless erections	Painful in the active phase
Other findings	Orthotopic meatus, complete foreskin, normal or above-average length	Trauma history; PD

Types of Congenital Curvature of the Penis

- **Embryology:** a ventral midline epithelial groove deepens and its edges **fuse proximal to distal** to the glans tip; surrounding mesenchyme forms the **spongiosum, Buck's fascia, dartos and ventral skin**. Fetal testosterone → **5 α -reductase** → **dihydrotestosterone (DHT)**, acting on androgen-receptor cells in all layers.
- Maturation of these layers depends on the same growth factors, so the **urethra may form normally while mesenchyme is deficient** → dysgenetic, inelastic fascia.

DEVINE AND HORTON CLASSIFICATION OF CONGENITAL CURVATURE (1973)

Type	Abnormal tissue	Notes
I	Corpus spongiosum deficient at maximal curvature	Types I–III = chordee without hypospadias (hypospadias anomaly)
II	Normal urethra and spongiosum; dysgenetic band of Buck's and dartos fascia → ventral curvature	
III	Dartos only ; may accompany a prominent suprapubic fat pad	
IV	Urethra, spongiosum and fascia normal; asymmetric compliance of the corporal tunica	= congenital curvature of the penis ; noticed before puberty, worsens with the pubertal growth spurt ; named by direction (ventral, lateral left or right)
V	Congenital short urethra : all layers fused correctly, but urethra and spongiosum not long or compliant enough on erection	So rare its existence should be doubted ; correction grouped with I–III; only the most experienced surgeons

- **Rule: if the meatus has reached the tip of the glans, do not divide the urethra** to correct ventral curvature; any apparent exception should itself be questioned.

Chordee Without Hypospadias in Young Males

- Ventral curvature, or **ventral curvature with torsion (complex)**; lifelong, may worsen at puberty; **average stretched length (13.1 cm)**.
- **Ventral skin abnormalities**: hooded preputial element or **high penoscrotal junction**; dorsal wrinkling despite a fused prepuce; **palpable ventral inelasticity on stretch** — dysgenetic tissue replacing Buck's and dartos fascia, sometimes tunical inelasticity.
- **Tissue studies**: **5 α -reductase deficiency** in ventral dysgenetic tissue; **fewer androgen receptors** in hypospadiac urethral mucosa; timing of neonatal androgen-receptor inactivation determines the anomaly; growth-factor deficiency correlates with hypospadias (not studied in chordee without hypospadias).

Erect photograph	Diagnosis
Erection commensurate with flaccid size	Chordee without hypospadias
Erection noticeably large	Congenital curvature of the penis

- Home photographs of the erect penis guide planning; **address psychological aspects**. Correction is highly successful — **single operation in almost all**.

- **The ventral corpora, not the spongiosum, usually limit straightening** — even with poor ventral fusion or a bifid spongiosum, and even where the distal urethra is only an epithelial tube. **If that tube is not stenotic, avoid dividing it.**

Operative sequence (children).

1. **Excise all ventral dysgenetic tissue and widely mobilise the spongiosum** from glans to perineum.
2. Residual chordee → under **artificial erection, midline ventral septotomy**: longitudinal sharp incision in the ventral midline, extended between the corpora so the ventral tunical edges move laterally.
3. Still curved → mobilise the **dorsal neurovascular structures with Buck's fascia**, excise **small dorsal tunical ellipse(s)** and close with **watertight plicating sutures**. **Caution**: with poor ventral development the **dorsal arteries may be the dominant supply to the glans**.

Congenital Curvatures of the Penis

- Ventral, **lateral (most often left)**, or unusually dorsal; a **smooth curve of the whole pendulous shaft**. Healthy men of **18–30 years** who noticed it before puberty, assumed it normal, and discovered otherwise with puberty or sexual activity; may increase through puberty.

Acquired Curvatures of the Penis

- Inevitably follow **trauma**: many are PD (attributed to microtrauma during intercourse; see Chapter 76); some follow **vigorous internal urethrotomy** extending through spongiosum into the corporal tunica.

Acquired Curvatures of the Penis From Causes Other Than Peyronie Disease

- A young man with acquired curvature → always consider PD, but many recall a **lateral buckling injury during intercourse**.

Entity	History	Lesion and course
Penile fracture	Snap, immediate detumescence, ecchymosis	Usually repaired acutely; if ignored → lateral indenting scar ± curvature
Subclinical fracture	Buckling without detumescence or ecchymosis ; painful erections, then a lateral nodule	Outer longitudinal tunical layer torn, inner circular intact (or both torn under intact Buck's fascia) → lateral linear scar, indentation, curvature
Septal insertion tear	Pop, painful erections	Curvature, usually dorsal ; mimics PD

- **Distinguished from PD** by usually **normal erectile function**, no global cavernosal veno-occlusive dysfunction, and **no penile shortening** — although some patients do develop erectile dysfunction (ED) after fracture-type injury.
- **Not candidates for contralateral plication**: it creates a second scar and a **near-circumferential indentation**, which upsets most patients despite a straight penis. Instead, **excise the scar and graft the corporotomy defect**; the lateral site needs minimal mobilisation of Buck's fascia, neurovascular bundle and spongiosum.

KEY POINTS: CURVATURES OF THE PENIS

- Chordee without hypospadias is a **forme fruste of hypospadias** (malformed ventral structures, erection not large); congenital curvature reflects **non-symmetrical expansion of the erectile bodies** and may come with above-average length.

Concept — Congenital curvature is either a **ventral mesenchymal deficiency on the hypospadias spectrum** (normal-sized erection) or **tunical asymmetry** (large erection); in both, the corpora rather than the urethra

usually limit straightening, so a urethra that reaches the glans tip is preserved. Post-traumatic lateral scars cause no shortening, so they are **excised and grafted, not plicated**.

PENILE TRANSPLANTATION

General

- After devastating phallic loss (trauma, malignancy, congenital anomaly) the options are **phalloplasty** (now common in masculinising gender-affirmation surgery; Chapter 128) or **vascularised composite allotransplantation (VCA)**, which is uncommon and not widely available.
- **VCA advantages:** natural appearance, **potential erectile function and normal voiding**. **Challenges:** **lifelong immunosuppression** and emotional, cultural, personal and ethical issues — of four successful transplants, **one was explanted on postoperative day 14 for psychological reasons**.

BURIED PENIS AND FOREIGN BODY INJECTION

Buried Penis

- **Adult acquired buried penis (AABP):** entrapment of the phallus in surrounding skin and/or soft tissue. **Obesity is the main risk factor;** also aggressive circumcision, scrotal lymphoedema, LS.
- **Pathogenesis:** excess prepubic tissue (**escutcheon**) → penis tethered to Scarpa fascia by the suspensory ligament → penopubic and penoscrotal angles migrate distally and the phallus telescopes → **trapped urine and moisture → inflammation and infection → scarring and LS → further trapping**.
- Consequences: impaired voiding and sexual function, **UTI, penile cancer**, poor mental health, avoidance of social and romantic situations. **Weight loss generally does not solve it.**

CLASSIFICATIONS OF ADULT ACQUIRED BURIED PENIS

System	Basis	Categories
Flynn	Seven anatomic features	Abdomen/pannus; mons/escutcheon; penile skin; scrotal skin; mons sulcus; pubic sulcus; scrotal sulcus (penoscrotal angle)
Wisconsin (Hesse)	Penile skin quality; pannus or scrotal lymphoedema	Four types, each with a treatment paradigm
Pariser	Reconstructive need	I local skin flap unburying; II skin graft; III scrotoectomy or scrotoplasty; IV escutcheonectomy; V abdominal panniculectomy
Higuchi	"Concealed" penis	Trapped (phimotic cicatrix); buried (no cicatrix); complex buried

- Burying from **LS or phimotic cicatrix only → trial of topical steroid**; otherwise AABP is a **surgical disease**.

Preoperative assessment.

- Examine and photograph **supine and standing** to judge pannus contribution; note scrotal, perineal and penile skin quality, cicatrix and LS. **Plastic surgery** if panniculectomy needed; **computed tomography (CT)** if hernia cannot be excluded clinically and panniculectomy is planned.
- **Evaluate the urethra** — high rate of stricture: dilation or meatotomy at repair, or staged urethroplasty. **Severe penile stricture → urethroplasty before any skin grafting**.
- Optimise diabetes; consider medical or surgical weight-loss referral; clarify goals, baseline erectile and voiding status, and **realistic cosmetic expectations**.

Operative sequence.

1. **Liberate the glans:** dorsal and/or ventral slit through the phimotic ring → glans traction stitch → circumcising incision → excise all nonviable skin, **preserving shaft-base skin** for the penopubic angle → measure the defect.
2. **Escutcheonectomy:** mark just below the abdomen–pannus sulcus, parallel to the inguinal folds, and as an **inverted U above the penile base**, keeping the penopubic angle and **2–3 cm of healthy skin at the shaft base** (otherwise tack a rim of healthy skin to the base); confirm tension-free apposition; **harvest any graft from the escutcheon before incising** → dissect to fascia, protecting spermatic cords and shaft, with a **vessel-sealing device** (LigaSure) for engorged veins → **tack Scarpa's superior edge to rectus fascia or pubis** either side of the penis → **two Jackson-Pratt drains** → superficial layer → skin with staples, interrupted mattress or running subcuticular barbed suture.
3. **Coverage:** STSG from thigh; FTSG or STSG from the resected escutcheon; local scrotal flap; or burial in a scrotal tunnel as a two-stage approach.
4. **STSG:** pneumatic dermatome at **0.012–0.020 inch**; on the escutcheon, penetrating towel clamps at four corners and liberal mineral oil → secure proximal and distal margins with interrupted and running **4-0 chromic** → fibrin glue beneath, then compress out excess → **bolster for 5–7 days** (or wound vacuum); some favour 1–5 days' bed rest. Thigh donor site: silver-impregnated adhesive dressing reinforced with gauze and Tegaderm, left to fall off.
5. **Aftercare:** dressing off at about 5 days; **drains out below 30 mL/day**; daily petroleum gauze at home for 2 more weeks. **Wound complications are common but mostly managed non-operatively.**

Scrotal Lymphedema

- **Distinct from obesity- or LS-related burying.** The phallus is buried in massive scrotal lymphoedema and voiding occurs through a **skin tunnel from the inferior scrotum**.
- **Resect all involved tissue;** perineal and lateral scrotal skin is often relatively spared and used for coverage. Start the incision over the **expected penile base**; identify and protect shaft and cords. Cords are **elongated** by years of scrotal weight — fold and tack them. A vessel-sealing device controls the **enlarged vessels and lymphatics** (Fig. 85.15).

BOX 85.1 CAUSES OF SCROTAL LYMPHEDEMA

Group	Causes
Congenital	Milroy disease (infancy); Meige disease (puberty); lymphoedema tarda (adult onset, underdeveloped lymphatics)
Infection	Lymphogranuloma venereum, <i>Wuchereria bancrofti</i> , streptococcal lymphangitis, tuberculosis, syphilis, leprosy, chancroid
Chronic inflammation	Hidradenitis suppurativa
Granulomatous	Sarcoidosis, rheumatoid arthritis, Crohn disease
Neoplastic	Inguinal or pelvic lymphadenopathy
Iatrogenic	Prior lymph node dissection, radiation
Reactive	Venous thrombosis, angioneurotic scrotal oedema
Fluid imbalance	Congestive heart failure
Idiopathic	—

Modified from McDougal WS, J Urol 2003.

- The table separates **congenital (three eponyms distinguished only by age of onset)** from a long acquired list; the acquired causes group into infective, inflammatory, granulomatous, neoplastic,

iatrogenic, reactive and fluid-balance categories, with *Wuchereria bancrofti* (filariasis) the classic infective cause.

Foreign Body Injection

- Self-injection for **self-perceived inadequate size**, across cultures since at least the late 1800s; physicians trialled silicone injection in the early 1900s before complications were appreciated. Agents: **silicone, paraffin, mercury, fat, petroleum jelly**.
- **Complications**: deformity, **skin necrosis, ED, abscess, cellulitis, Fournier gangrene, pulmonary embolism, pneumonitis**. Silicone is inert but triggers **fibrosis and collagen deposition months to decades later**; enlargement can preclude intercourse or concealment.
- **Active infection** → admission, intravenous antibiotics, drainage of abscesses, debridement of necrotising infection. **No infection but wishes normal size** → **excise all affected tissue and skin**, cover with scrotoplasty and STSG (Fig. 85.16).
- **Body dysmorphic disorder** is common and most have **average length and girth** → consider **psychiatry referral**; counsel frankly on realistic outcomes.
- **Penuma**: surgically implanted **silicone sheath** adding girth and length to the flaccid penis; small series report satisfaction and length gain, but it is controversial — long-term satisfaction uncertain and complications (**scarring, erosion**) possibly under-reported.

KEY POINTS: BURIED PENIS AND FOREIGN BODY INJECTION

- Buried penis arises from obesity, LS, scrotal lymphoedema or a combination; **rarely cured by weight loss alone**; trapped urine drives skin loss that often **requires grafting**.
- Injected augmentation material causes a **delayed, often dramatic fibrotic reaction**.

Concept — Buried penis is a **self-perpetuating cycle** (escutcheon → telescoping → trapped urine → inflammation and LS → more trapping), so weight loss alone fails and repair must **remove the burying tissue, restore the angles and replace lost skin**; lymphoedema and injected material obey the same rule — excise all diseased tissue, then resurface.